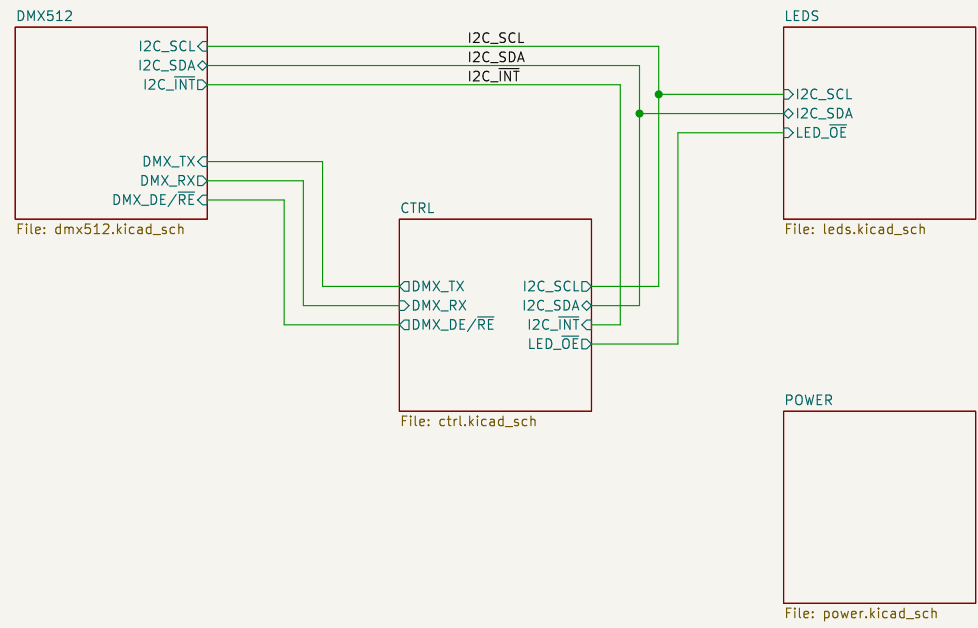


LED Beacon

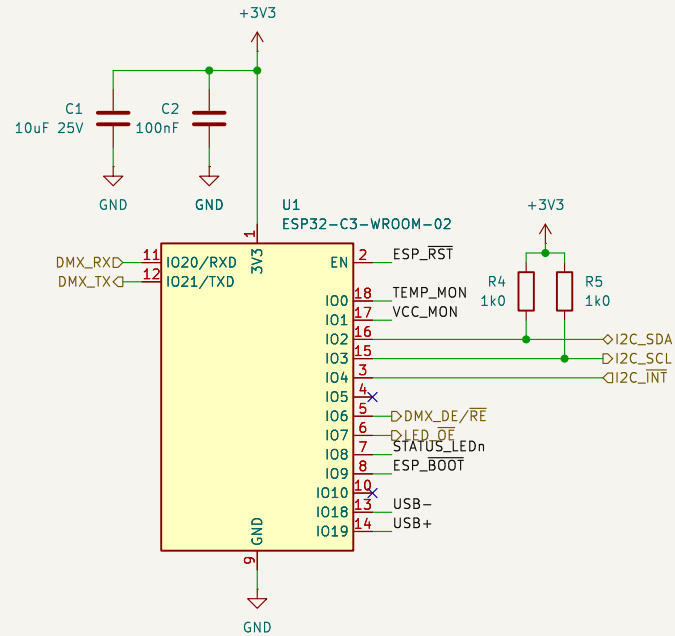
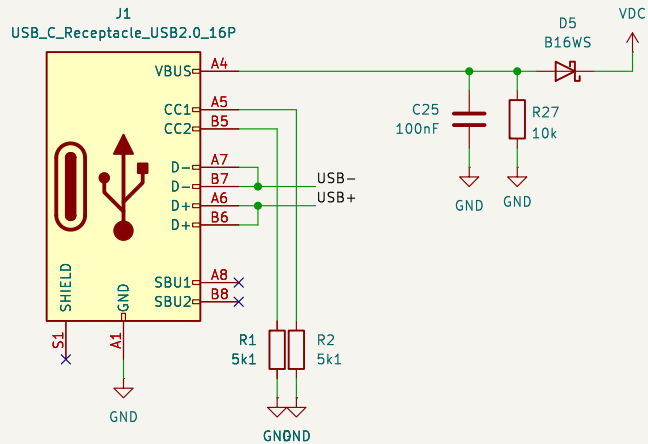


- H1 MountingHole
- H2 MountingHole
- H3 MountingHole
- H4 MountingHole

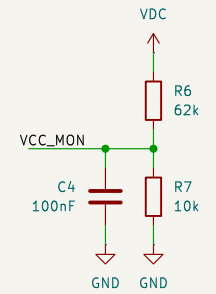
Wires & Bits		
Sheet: /		
File: emf_beacon.kicad_sch		
Title: LED Beacon		
Size: A4	Date: 2026-06-05	Rev: 1
KiCad E.D.A. 9.0.9		Id: 1/5

Controller

USB Programming

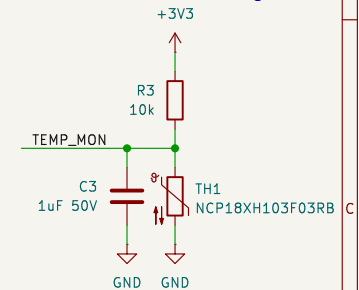


VCC Monitoring

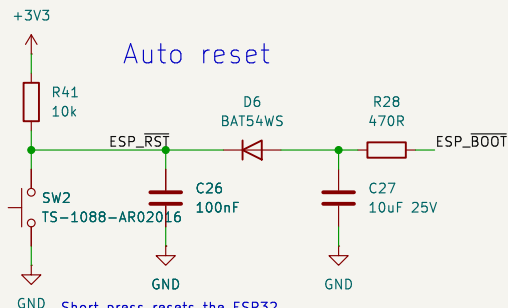


Voltage Monitor
 V_{max} is 17.5V
 $V_{adc} \times ((43k\Omega + 10k\Omega) + 10k\Omega) = VCC$
 $V_{adc} \times 5.3 = VCC$
 $1lsb = 4.271\mu V$

Temperature Monitoring

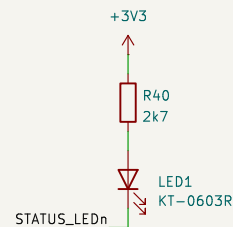


Auto reset



Short press resets the ESP32
 Long press resets the ESP32
 and puts it in bootloader mode

Status LED



I08 needs to be held high to boot
 so LED & resistor are high side.
 Set STATUS_LEDn to 0V to light LED.

Wires & Bits

Sheet: /CTRL/
 File: ctrl.kicad_sch

Title: Controller

Size: A4 Date: 2026-06-05
 KiCad E.D.A. 9.0.9

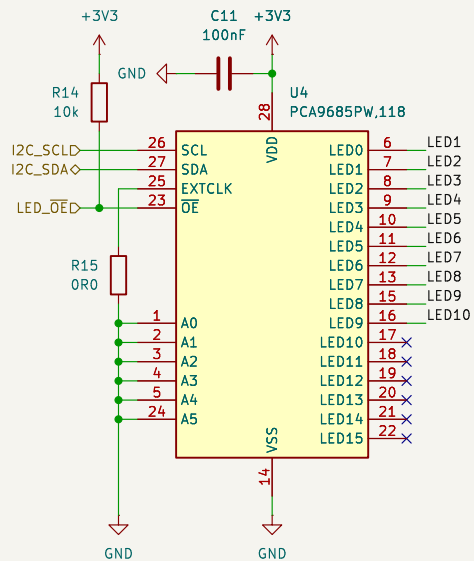
Rev: 1
 Id: 2/5

LED Driver

LED Enable

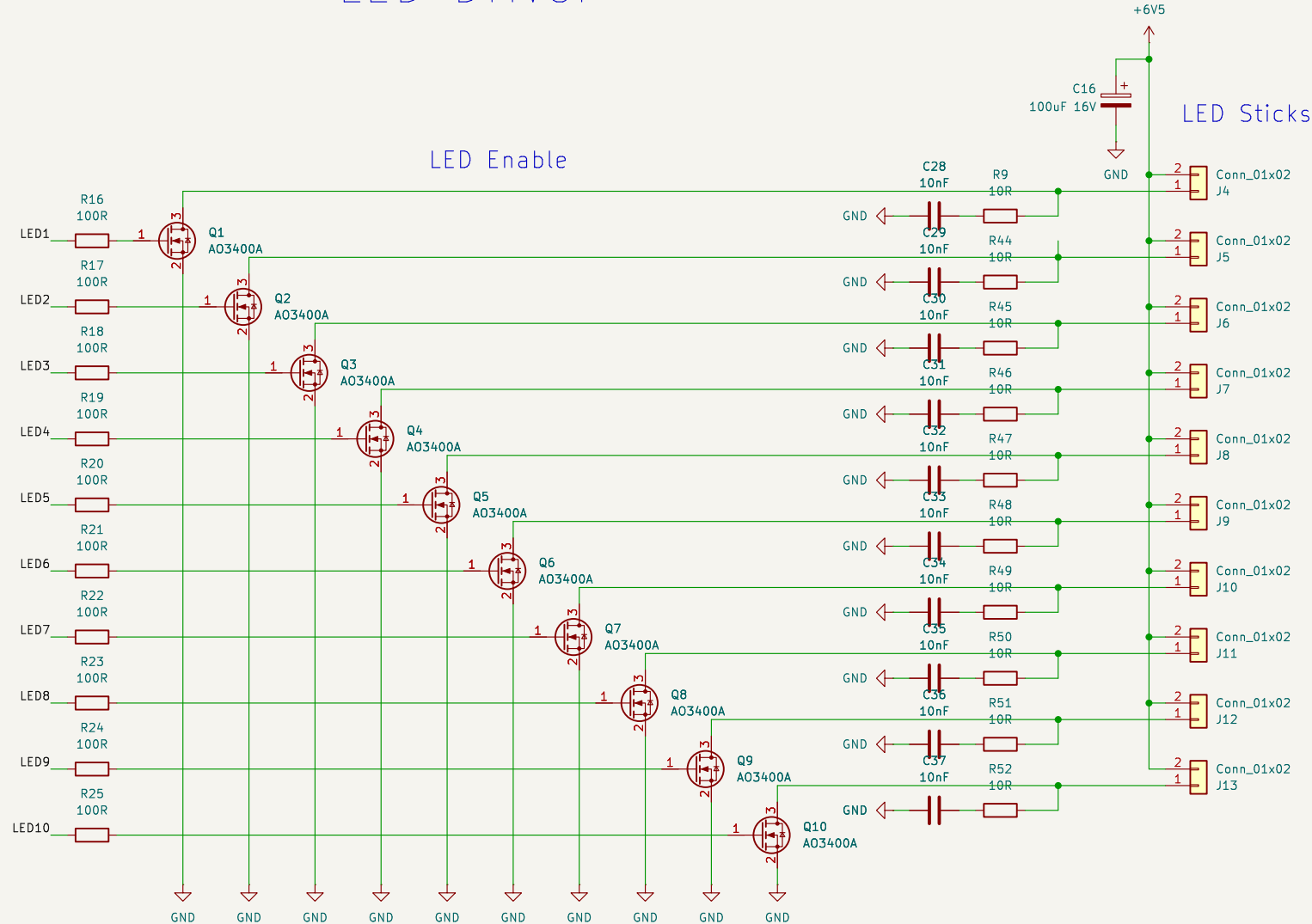
LED Sticks

I2C → PWM



I2C Addr: 0x40

A string of 6x LEDs is 6.4Vf @ 0.3A



Wires & Bits

Sheet: /LEDS/
File: leds.kicad_sch

Title: LEDs

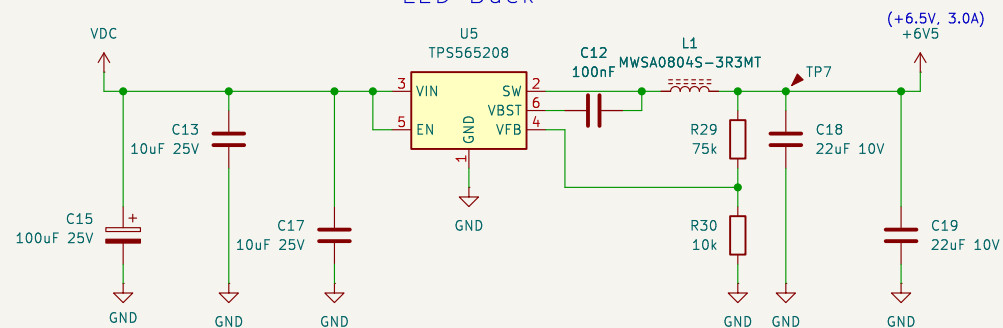
Size: A4
KiCad E.D.A. 9.0.9

Date: 2026-06-05

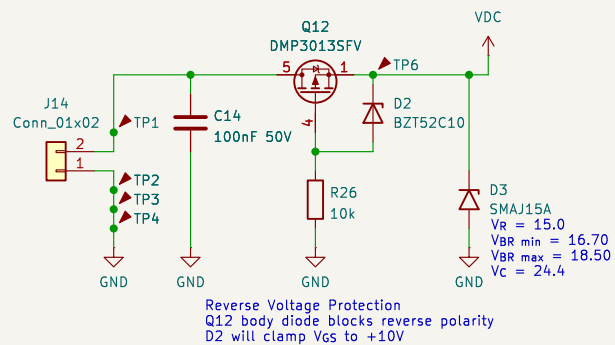
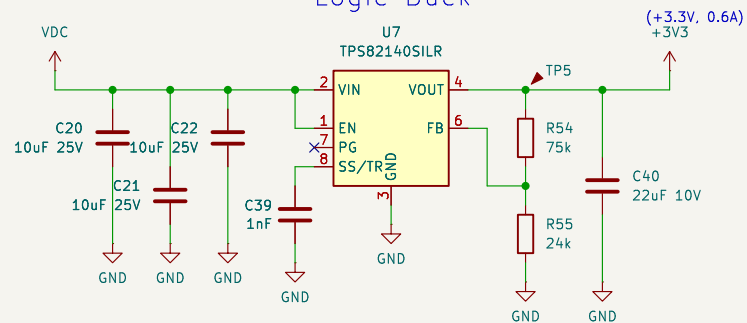
Rev: 1
Id: 4/5

Power Supplies

LED Buck



Logic Buck



EMC ferrites /CMC?

Wires & Bits

Sheet: /POWER/
File: power.kicad_sch

Title: Power

Size: A4 Date: 2026-06-05
KiCad E.D.A. 9.0.9

Rev: 1
Id: 5/5